

Database Structure Documentation

HabitTrackerAspNetMvcWebApp

1. Overview

The application uses Entity Framework Core (Code-First approach) with a SQL Server database. The schema is defined through C# model classes in the **Models** folder and configured via **ApplicationDbContext** (in the **Data** folder). The database consists of two application-specific tables (**Habits** and **HabitLogs**) plus the standard ASP.NET Core Identity tables used for user authentication and authorization (Identity was extended with a custom **ApplicationUser** class). No diagram is included below; each table's structure, keys, and relationships are described in text and tables instead.

Tables in the database:

Table	Purpose
AspNetUsers	Application users (extends ASP.NET Identity user, adds IsActive)
AspNetRoles	Identity roles
AspNetUserRoles	Join table: users ↔ roles
AspNetUserClaims	Claims associated with a user
AspNetRoleClaims	Claims associated with a role
AspNetUserLogins	External login providers linked to a user
AspNetUserTokens	Tokens associated with a user (e.g. 2FA, password reset)
Habits	Habits created by a user (application-specific)
HabitLogs	Daily log entries recorded against a habit (application-specific)

The two core application-specific tables — **Habits** and **HabitLogs** — are documented in full detail in Section 2. The ASP.NET Identity tables are documented in Section 3.

2. Application-Specific Tables

2.1 Habits

Stores each habit created by a user. Defined by the **Habit** model class and mapped to the **Habits** table.

Column	Type	Constraints	Description
Id	int	Primary Key, Identity (auto-increment)	Unique habit identifier
Title	nvarchar(100)	Required	Habit title (max 100 characters)
Description	nvarchar(500)	Optional	Habit description (max 500 characters)
Frequency	int (enum)	Required	Daily = 0, Weekly = 1, Monthly = 2
Status	int (enum)	Required	Active = 0, Completed = 1, Paused = 2
KanbanStatus	int (enum)	Required, default Todo	Todo = 0, InProgress = 1, Done = 2
StartDate	datetime2	Required	Date the habit begins
EndDate	datetime2	Optional	Date the habit ends (nullable)
UserId	nvarchar(450)	Required, Foreign Key → AspNetUsers.Id	Owner of the habit

Indexes: non-unique index on UserId.

Relationships: Many Habits belong to one ApplicationUser (UserId → AspNetUsers.Id, ON DELETE CASCADE). One Habit has many HabitLogs.

2.2 HabitLogs

Stores individual log entries recorded for a habit on a given date. Defined by the **HabitLog** model class and mapped to the **HabitLogs** table.

Column	Type	Constraints	Description
Id	int	Primary Key, Identity (auto-increment)	Unique log entry identifier
HabitId	int	Required, Foreign Key → Habits.Id	The habit this entry belongs to
LogDate	datetime2	Required	The calendar date of the log entry
Status	int (enum)	Required, default Completed	Completed = 0, PartiallyCompleted = 1, Skipped = 2

Indexes: unique composite index on (HabitId, LogDate) — a habit can only have one log entry per date.

Relationships: Many HabitLogs belong to one Habit (HabitId → Habits.Id, ON DELETE CASCADE).

3. ASP.NET Core Identity Tables

These tables are provided by ASP.NET Core Identity (IdentityDbContext) and handle user accounts, roles, and authentication. The user table is extended by the custom **ApplicationUser** class, which adds an **IsActive** flag.

3.1 AspNetUsers (ApplicationUser)

Column	Type	Constraints	Description
Id	nvarchar(450)	Primary Key	Unique user identifier (string GUID)
UserName	nvarchar(256)	—	Login username
NormalizedUserName	nvarchar(256)	Unique index (UserNameIndex)	Uppercased username, used for lookups
Email	nvarchar(256)	—	User's e-mail address
NormalizedEmail	nvarchar(256)	Index (EmailIndex)	Uppercased e-mail, used for lookups
EmailConfirmed	bit	—	Whether the e-mail has been confirmed
PasswordHash	nvarchar(max)	—	Hashed password
SecurityStamp	nvarchar(max)	—	Changes when credentials change
ConcurrencyStamp	nvarchar(max)	Concurrency token	Used to detect concurrent updates
PhoneNumber	nvarchar(max)	—	User's phone number
PhoneNumberConfirmed	bit	—	Whether the phone number is confirmed
TwoFactorEnabled	bit	—	Whether 2FA is enabled
LockoutEnd	datetimeoffset	Optional	Date/time the lockout ends, if any
LockoutEnabled	bit	—	Whether lockout is enabled for the user
AccessFailedCount	int	—	Number of failed login attempts
IsActive	bit	Custom field, default true	Whether the account is active (added by the project)

Relationships: referenced by Habits.UserId, AspNetUserRoles, AspNetUserClaims, AspNetUserLogins, AspNetUserTokens (all cascade delete).

3.2 AspNetRoles

Column	Type	Constraints	Description
Id	nvarchar(450)	Primary Key	Unique role identifier
Name	nvarchar(256)	—	Role name
NormalizedName	nvarchar(256)	Unique index (RoleNameIndex)	Uppercased role name
ConcurrencyStamp	nvarchar(max)	Concurrency token	Used to detect concurrent updates

3.3 AspNetUserRoles

Join table linking users to roles (many-to-many).

Column	Type	Constraints	Description
UserId	nvarchar(450)	Primary Key (composite), FK →AspNetUsers.Id	The user
RoleId	nvarchar(450)	Primary Key (composite), FK →AspNetRoles.Id	The role

3.4 AspNetUserClaims

Column	Type	Constraints	Description
Id	int	Primary Key, Identity	Unique claim identifier
UserId	nvarchar(450)	Required, FK →AspNetUsers.Id	The user the claim belongs to
ClaimType	nvarchar(max)	—	Type of the claim
ClaimValue	nvarchar(max)	—	Value of the claim

3.5 AspNetRoleClaims

Column	Type	Constraints	Description
Id	int	Primary Key, Identity	Unique claim identifier
RoleId	nvarchar(450)	Required, FK →AspNetRoles.Id	The role the claim belongs to
ClaimType	nvarchar(max)	—	Type of the claim
ClaimValue	nvarchar(max)	—	Value of the claim

3.6 AspNetUserLogins

Stores external login provider information linked to a user account.

Column	Type	Constraints	Description
LoginProvider	nvarchar(128)	Primary Key (composite)	Name of the external login provider
ProviderKey	nvarchar(128)	Primary Key (composite)	Key given by the provider
ProviderDisplayName	nvarchar(max)	—	Display name of the provider
UserId	nvarchar(450)	Required, FK → AspNetUsers.Id	The linked user

3.7 AspNetUserTokens

Stores tokens for a user, such as those used for 2FA or password reset.

Column	Type	Constraints	Description
UserId	nvarchar(450)	Primary Key (composite), FK → AspNetUsers.Id	The user the token belongs to
LoginProvider	nvarchar(128)	Primary Key (composite)	Provider associated with the token
Name	nvarchar(128)	Primary Key (composite)	Token name
Value	nvarchar(max)	—	Token value

4. Relationships Summary

Foreign Key	References	Cardinality	On Delete
Habits.UserId	AspNetUsers.Id	Many-to-One	Cascade delete
HabitLogs.HabitId	Habits.Id	Many-to-One	Cascade delete
AspNetUserRoles.UserId	AspNetUsers.Id	Many-to-One	Cascade delete
AspNetUserRoles.RoleId	AspNetRoles.Id	Many-to-One	Cascade delete
AspNetUserClaims.UserId	AspNetUsers.Id	Many-to-One	Cascade delete
AspNetRoleClaims.RoleId	AspNetRoles.Id	Many-to-One	Cascade delete
AspNetUserLogins.UserId	AspNetUsers.Id	Many-to-One	Cascade delete
AspNetUserTokens.UserId	AspNetUsers.Id	Many-to-One	Cascade delete

In plain terms: one **ApplicationUser** can own many **Habits**; one **Habit** can have many **HabitLogs** (at most one per calendar date, enforced by a unique composite index on HabitId + LogDate); and one **ApplicationUser** can have many roles, claims, external logins, and tokens through the standard Identity join/support tables.